ELECTRICAL—ADDING ADDITIONAL BULBS TO TURN INDICATOR OR HAZARD FLASHER CIRCUITS—SERVICE TIPS	Article No. 02-21-9

FORD: 1999-2003 SUPER DUTY F SERIES
2000-2003 EXCURSION, F-650, F-750

Article **01-16-8** is being republished in its entirety to update the model line and model year coverage.

ISSUE

When adding additional bulbs to the turn signal or hazard flasher circuits, isolation relays must be used in order to prevent damage to the flasher relay module and or the multi function switch, and to prevent erratic operation.

ACTION

The following current ranges must be adhered to on the original flasher circuitry:

- Minimum rated current for the turn signal (either left or right) is 4.45A, at 12.8 V and the maximum rated current is 5.06A at 12.8 V
- Turn signal current draw below 2.83A at 12.8V will cause the flasher to flash rapidly
- The minimum rated current for the hazard flasher is 4.20A at 12.8 V and the maximum rated current is 10.12A at 12.8 V

Any additional current load above the maximum rating must use isolation relays. Refer to Figure 1 for typical relay installation illustration.

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 01-16-8
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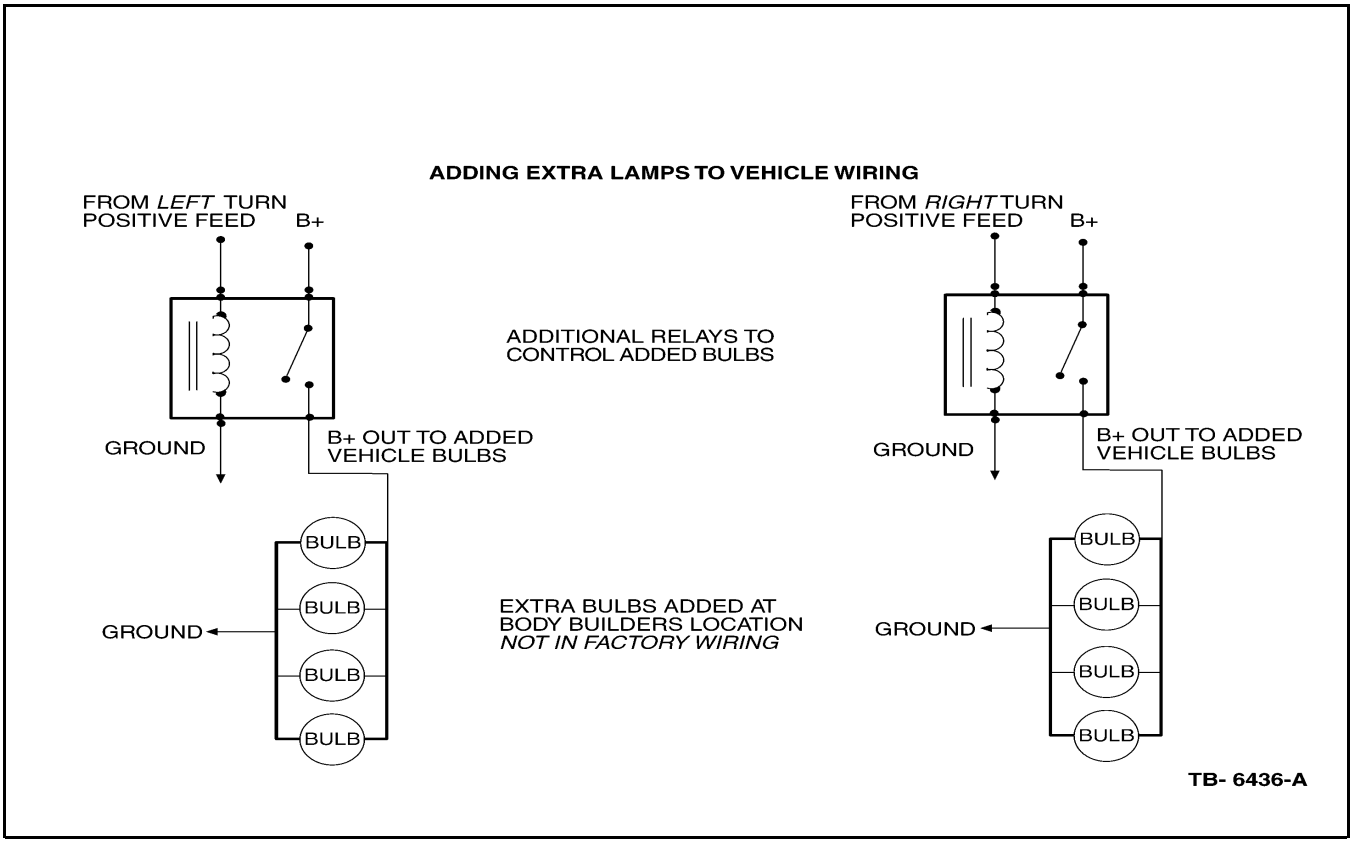

Figure 1 - Article 02-21-9

NOTE: The information in Technical Service Bulletins is intended for use by trained, professional technicians with the knowledge, tools, and equipment to do the job properly and safely. It informs these technicians of conditions that may occur on some vehicles, or provides information that could assist in proper vehicle service. The procedures should not be performed by "do-it-yourselfers". Do not assume that a condition described affects your car or truck. Contact a Ford, Lincoln, or Mercury dealership to determine whether the Bulletin applies to your vehicle.